UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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|---|---|
| COURSE | COMPUTER ORGANIZATION AND ARCHITECTURE |
| COURSE CODE | CSC429 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 6 printed pages.*

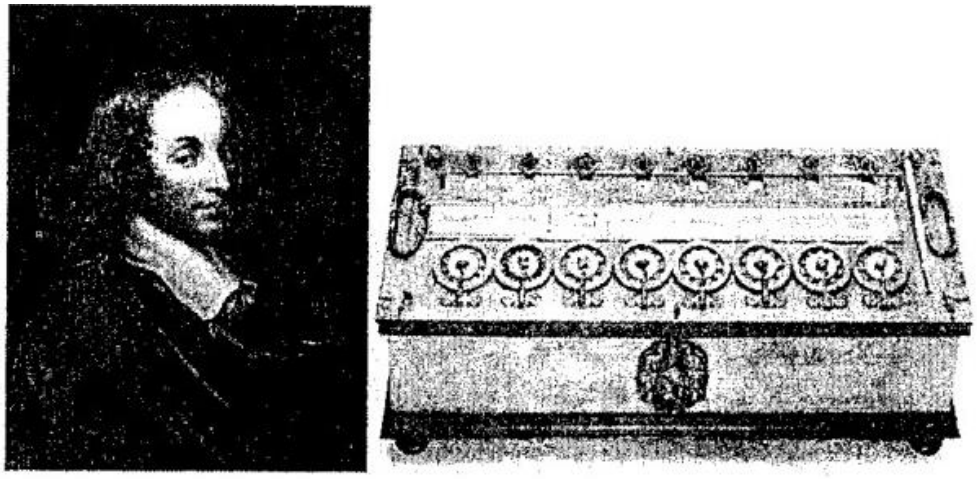
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**PART A (80 MARKS)**

Answer ALL questions in this part.

**QUESTION A1** **(5 marks)**

Question 1 is based on the following Figure 1.



Inventor 'Y' was France's most celebrated mathematician and physicist. In 1642, inventor 'Y' invented and build the first working calculating machine as a means of helping his father to perform tedious tax accounting.

a) Who is inventor 'Y'?

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

c) Which programming language is named after inventor 'Y'?

d) State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

**QUESTION A2** **(5 marks)**

Calculate -31 - (-12) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

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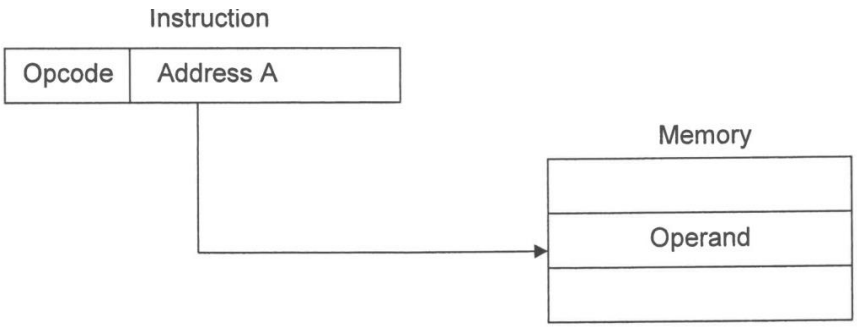
**QUESTION A3** **(5 marks)**

a) How classification of parallel processor is constructed?

b) Give ONE (1) example for each parallel processor architecture.

**QUESTION A4** **(5 marks)**

Given the following Figure 2.




a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

**QUESTION A5** **(10 marks)**

a) Describe THREE (3) types of interfaces to external devices.

b) Explain TWO (2) types of interrupt.

**QUESTION A6** **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert -6.75 to IEEE 754 single precision. Give the answer in hexadecimal.

b) Convert the IEEE 754 single-precision value 0x42C90000 to decimal.

**QUESTION A7** **(10 marks)**

a) When is an I/O module installed?

b) What will happen if the I/O module is not installed?

c) Explain ONE (1) category of I/O module

d) What does it mean by 'hot connection'?

e) List THREE (3) common interfaces to external devices.

**QUESTION A8** **(10 marks)**

a) Differentiate between organization and architecture.

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

**QUESTION A9** **(10 marks)**

a) Differentiate SRAM and DRAM from the perspective of the following characteristics,

i. The way the bits are stored.

ii. Speed.

iii. Location it is being used.

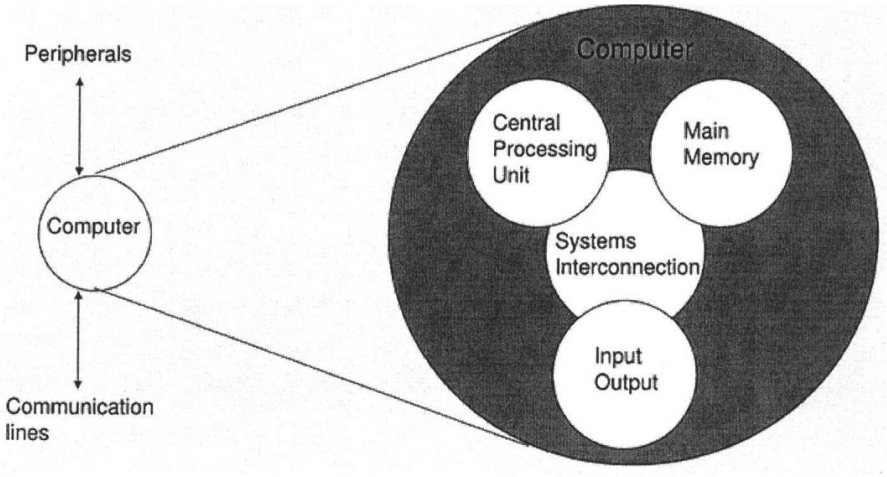
b) Why does DRAM need refreshing?

c) Comparing DRAM and SRAM, which memory would be placed higher in the memory hierarchy? Give your justification to support your answer.

**QUESTION A10** **(10 marks)**

Computer consists of four main components as illustrated in Figure 1. Each component has its own function in ensuring that the computer system works well. Question 4 is based on

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a) Explain THREE (3) types of system interconnection.

b) State TWO (2) sub-components of central processing unit and its function.

**PART B (20 MARKS)**

Answer the question in this part.

**QUESTION B1** **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A(A + 2B / C - D^2)$$

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.

| Zero-address | One-address | Three-address |
|---|---|---|
| PUSHp | LOADp | ADD (p,q,r) |
| POPp | STORp | SUB (p,q,r) |
| ADD | ADDp | MUL (p.q.r) |
| SUB | SUBp | DIV(p,q,r) |
| MUL | MULp | |
| DIV | DIVp | |

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